

GhostNet-AI — Trustworthy Sonar Intelligence

SIH26057 • Research Challenges, Innovation Modules & Proposed Architecture

Yes. I looked at **SIH26057 itself plus recent 2025–2026 SSS research and publicly visible SIH26057 implementations**. The important finding is that the obvious solution space is already getting crowded: YOLO detection, confidence scoring, geotagging, dashboards, edge inference, and human review are already appearing in public work. ([GitHub])(https://github.com/Rehan9599/Sonar-Drishti?utm_source=chatgpt.com "GitHub - Rehan9599/Sonar-Drishti · GitHub"))

So, to make **GhostNet-AI** genuinely different, I would not try to add 20 normal features. I would attack the **hardest unresolved problems of real-world sonar deployment**.

The biggest challenge is not:

"Can AI find a box around a sonar object?"

It is:

***"Can the system remain trustworthy when the sonar, environment, object, location and training data are all different from what it saw during development?"**

That is where I would build your innovation.

1. BIGGEST CHALLENGE: AI works on one sonar, fails on another

This is one of the most serious real-world problems.

SSS images can vary because of:

- different sonar manufacturers
- frequency
- range
- altitude
- seabed type
- water conditions
- noise
- survey platform
- acquisition settings

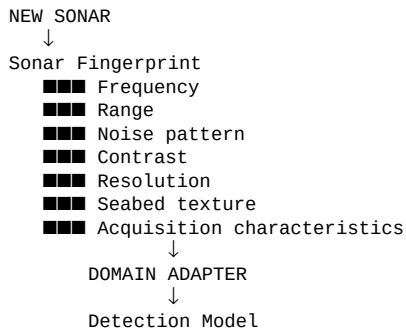
Recent research explicitly treats **domain shift / cross-domain generalization** as a major limitation; models trained in one environment can degrade in previously unseen environments. ([ResearchGate])(https://www.researchgate.net/publication/389936412_Contrastive_Adaptation_on_Domain_Augmentation_for_Generalized_Zero-Shot_Side-Scan_Sonar_Image_Classification?utm_source=chatgpt.com "Contrastive Adaptation on Domain Augmentation for Generalized Zero-Shot Side-Scan Sonar Image Classification | Request PDF"))

A 2026 domain-adaptive shipwreck paper still describes large domain shifts and limited samples as key obstacles to stable deployment.
([ScienceDirect](https://www.sciencedirect.com/science/article/abs/pii/S0029801825034535?utm_source=chatgpt.com "Effective domain adaptive shipwreck detection method in side-scan sonar images for unmanned underwater vehicles - ScienceDirect"))

Your solution

****GhostNet-AI Universal Sonar Adapter****

Before detection, your system estimates the current sonar domain:



The important part:

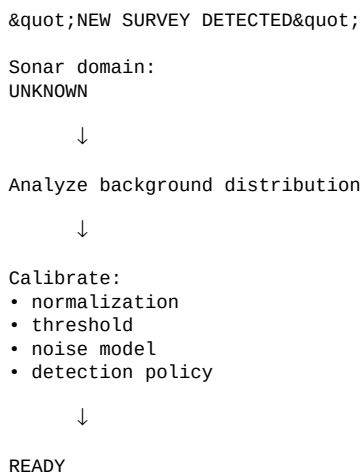
Don't retrain the entire model every time.

Adapt the input/features to the new sonar domain.

2. Make the system SELF-CALIBRATING

This can become one of your strongest innovations.

When a new survey is uploaded:



So the AI doesn't assume:

"Every sonar image looks like my training data."

Instead it says:

"Let me understand this sensor/survey first."

This is much stronger than just claiming "robust to noise."

3. BIGGEST DATA PROBLEM: real ghost-net labels are scarce

This is a serious limitation.

The public sonar ecosystem has many datasets for shipwrecks, aircraft and generic objects, but genuinely labelled ghost-net data is much harder to obtain. A 2025 survey of sonar datasets identifies dataset scarcity as a fundamental bottleneck, and recent public SIH26057 work also notes that real ghost-net SSS data are limited. ([IEEE Xplore](https://ieeexplore.ieee.org/document/11223279/?utm_source=chatgpt.com "Sonar Image Datasets: A Comprehensive Survey of Resources, Challenges, and Applications | IEEE Conference Publication | IEEE Xplore"))

GhostVision's recent work demonstrates that real-world ghost-gear detection can be done with low-cost SSS, but it also highlights operational issues such as false positives, threshold sensitivity and georeferencing uncertainty. ([MDPI](https://www.mdpi.com/2077-1312/14/10/951?utm_source=chatgpt.com "GhostVision: Democratizing Derelict Gear Detection Using Low-Cost Sonar and Artificial Intelligence | MDPI"))

Don't solve this with only synthetic images.

Instead build:

****Active Learning + Weakly/Self-Supervised Sonar Learning****

```
Large unlabeled sonar
↓
Self-supervised learning
↓
Sonar representation
↓
Small labeled ghost-net set
↓
Detection model
↓
UNCERTAIN EXAMPLES
↓
Human labels only difficult cases
↓
Retraining
```

Your AI actively asks:

"Which 50 sonar regions should the expert label?"

That's much more intelligent than randomly collecting more data.

4. HARD CHALLENGE: natural seabed looks like debris

This is directly central to your PS.

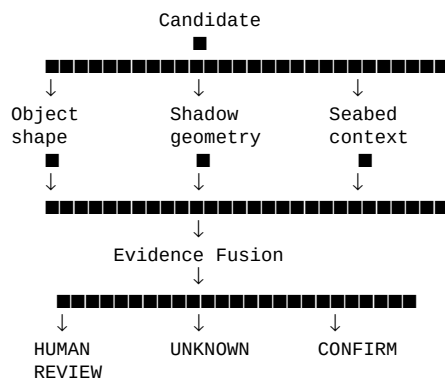
Natural rock formations, sand ripples, ridges and acoustic shadows can look like man-made objects. Recent SSS detection literature continues to report high false-alarm difficulty from background noise and natural structures. ([Frontiers s])(https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2025.1542832/full?utm_source=chatgpt.com "Frontiers | Small object detection in side-scan sonar images based on SOCA-YOLO and image restoration"))

And field ghost-gear surveys report that rocky substrates create especially difficult sonar clutter. ([General Oceans Foundation])(https://www.generaloceansfoundation.com/news/maine-ghost-gear-detection-with-side-scan-sonar?utm_source=chatgpt.com "Maine Ghost Gear Detection with Side-scan Sonar"))

Don't make YOLO the final judge.

Create:

****Anthropogenic Evidence Engine****



The key question becomes:

****"Is there enough independent evidence that this is anthropogenic?"****

That is much more meaningful than a raw YOLO score.

5. HARD CHALLENGE: AI confidence is not the same as truth

This is another place where your project can become substantially different.

Imagine:

YOLO confidence = 94%

But:

- sonar quality is poor
- target is at long range

- shadow is ambiguous
- target was seen only once

Should the system say:

"94% confirmed"?

No.

Your system should calculate:

****Operational Trust****

For example:

MODEL	
Confidence	94%
SONAR QUALITY	
Good	✓
SHADOW	
Strong	✓
TRACK	
11 frames	✓
DOMAIN MATCH	
Poor	■
LOCATION	
±18 m	■
FINAL STATUS	
REVIEW REQUIRED	

This makes:

model confidence $\neq$ operational trust

a core principle.

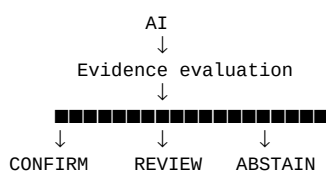
6. Build a “CONFIRM / REVIEW / ABSTAIN” AI

This is worth making an explicit feature.

Most systems force:

class = ghost net

Your system can say:



CONFIRM

Evidence is consistent.

REVIEW

Evidence conflicts.

ABSTAIN

Not enough evidence.

This is especially appropriate for underwater sonar because uncertainty comes from both sensing and environmental variation. GhostVision's field evaluation explicitly discusses false positives, threshold sensitivity and georeferencing uncertainty. ([MDPI](https://www.mdpi.com/2077-1312/14/10/951?utm_source=chatgpt.com "GhostVision: Democratizing Derelict Gear Detection Using Low-Cost Sonar and Artificial Intelligence | MDPI"))

7. BIG CHALLENGE: the same target appears in many images

This is often overlooked by simple demos.

A sonar survey is a sequence.

Frame 201 ■■
Frame 202 ■■
Frame 203 ■■
Frame 204 ■■
Frame 205 ■■

One physical ghost net may appear in all five.

A naive detector reports:

5 objects

Your system reports:

1 persistent target

Build:

****Target Identity / Sonar Tracking****

```
detections
↓
spatial-temporal association
↓
TARGET #017
  ■■■ 14 observations
  ■■■ average evidence
  ■■■ best sonar view
  ■■■ consolidated location
```

This is a very important transition:

Image intelligence → Survey intelligence

8. BIG CHALLENGE: exact location is harder than it looks

Your PPT already talks about location uncertainty.

Push this much further.

A detection isn't simply:

GPS = X,Y

Actual position can be affected by survey motion, range geometry and navigation uncertainty. Recent GhostVision work explicitly reports that georeferencing uncertainty grows with factors such as heading and slant-range conversion. ([MDPI](https://www.mdpi.com/2077-1312/14/10/951?utm_source=chatgpt.com "GhostVision: Democratizing Derelict Gear Detection Using Low-Cost Sonar and Artificial Intelligence | MDPI"))

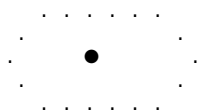
Build:

****Probabilistic Target Geolocation****

Instead of:

TARGET
●

show:



Estimated location
+
uncertainty region

Better:

Target center
●

95% estimated location region
■

And calculate the estimate using multiple observations:

Frame 1 → location estimate
Frame 2 → location estimate
Frame 3 → location estimate
...
↓
fused estimate
↓

uncertainty area

Now geolocation itself becomes part of the intelligence.

9. BIG CHALLENGE: what if the object is something the model has never seen?

This is the **open-world problem**.

Suppose the model knows:

Ghost Net
Wreck
Pipe
Cylinder
Aircraft

But finds:

????????????

A normal classifier will often force it into the nearest known class.

Instead create:

****UNKNOWN MAN-MADE ANOMALY****

Known object
↓
Known class

Unknown but suspicious
↓
Anthropogenic anomaly
↓
Human investigation

Then save the human label for future model improvement.

This is particularly relevant because generalized/zero-shot SSS recognition is still an active research problem; recent work specifically addresses never-seen targets and cross-domain generalization. ([ResearchGate](https://www.researchgate.net/publication/389936412_Contrastive_Adaptation_on_Domain_Augmentation_for_Generalized_Zero-Shot_Side-Scan_Sonar_Image_Classification?utm_source=chatgpt.com "Contrastive Adaptation on Domain Augmentation for Generalized Zero-Shot Side-Scan Sonar Image Classification | Request PDF"))

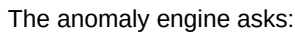
10. Combine ANOMALY DETECTION + OBJECT DETECTION

This is one of the approaches I think could make you stand out.

Most teams:

Image
↓

You:



The detector asks:

The fusion engine asks:

That gives you a fundamentally different architecture.

...

Don't assume an entire survey is equally good.

You can have:

Create:

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And dynamically alter the AI:

GOOD
→ normal mode

MODERATE
→ conservative mode

POOR
→ abstain more + human review

This directly addresses the PS's real-world acoustic difficulties and the documented effects of vessel motion and sonar quality. ([The Global Ghost Gear Initiative](https://www.ghostgear.org/aldfg-research-2024?utm_source=chatgpt.com "State of ALDFG Research 2024 — The Global Ghost Gear Initiative"))

12. BIG CHALLENGE: no standard way to compare models

This is surprisingly important.

A 2025 review notes the need for standardized benchmarks and common evaluation metrics for SSS computer vision. ([Liverpool Repository](https://livrepository.liverpool.ac.uk/3189677/?utm_source=chatgpt.com "Computer vision methods for side scan sonar imagery - The University of Liverpool Repository"))

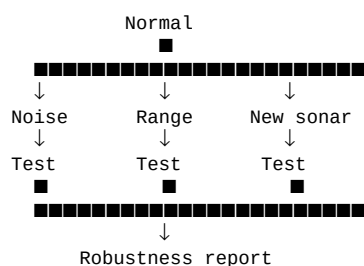
So you can create your own:

****GhostNet-AI Sonar Robustness Benchmark****

Don't just report:

mAP = X

Evaluate:



Measure:

- normal detection
- noisy detection
- low contrast
- unseen sonar
- unseen seabed
- long-range
- small object

- hard negatives

Then your claim becomes:

***"We don't only test whether the detector works. We test when it stops being trustworthy."**

That is a very strong research narrative.

13. BIG CHALLENGE: knowing when NOT to process

This connects beautifully to your AI-agent idea.

Suppose the agent sees:

```
XTF loaded
↓
GPS missing
↓
sonar quality poor
↓
domain mismatch high
```

Rather than blindly running everything:

****Agent decides the safe workflow****

Agent:

GPS missing.
Exact geolocation unavailable.

Sonar quality: moderate.

Recommended:
Run detection + flag location as uncertain.

Continue?
[YES] [CANCEL]

Now the agent is not merely clicking UI buttons.

It is **managing uncertainty**.

That is much more advanced.

14. The AI agent can become your SYSTEM ORCHESTRATOR

Your previous idea fits perfectly here.

Don't make the agent the "AI detector."

Make it:

****SonarOps Agent****

It controls the whole pipeline.

Operator:

****"Process Survey 014, find likely ghost gear, ignore low-quality regions, review ambiguous targets and generate the report."****

Agent:

```
PLAN
↓
Inspect XTF
↓
Check metadata
↓
Assess sonar health
↓
Select processing policy
↓
Detect
↓
Fuse evidence
↓
Track
↓
Geolocate
↓
Filter
↓
Request human review
↓
Generate report
```

This lets the operator focus on **exceptions**, rather than operating every button.

15. The REALLY ambitious problem: “Have we searched enough?”

This is bigger than detection.

Imagine:

AI found 7 targets

That doesn't tell you whether the rest of the area was properly searched.

Build:

****Survey Coverage Intelligence****

GREEN = Good coverage
YELLOW = Low-quality coverage
RED = Target
GRAY = No usable data

Then:

Survey coverage
+

Sonar quality
+
Target locations
↓
Remaining blind spots

The system could tell the operator:

“18% of the survey corridor has insufficient-quality coverage for reliable screening.”

That is a much more useful operational statement than “no object detected.”

...

16. Go one step further: RE-SCAN RECOMMENDATION

Now you have something genuinely interesting.

If a region is:

high importance
+
poor sonar quality

the system creates:

****Recommended Re-Survey Zone****

```
■ RE-SCAN RECOMMENDED ■  
■ Reason: poor coverage ■  
■ + suspected target ■  
■ Priority: HIGH ■
```

So the AI doesn't only say:

“I found something.”

It can say:

“This area needs another look.”

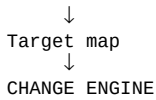
That is a much bigger problem-solving contribution.

...

17. Repeat-survey change intelligence

Then you can evolve that into:

SURVEY A
↓
Target map
↓
SURVEY B



Output:

```

NEW
REMOVED
PERSISTENT
UNCERTAIN
  
```

This moves GhostNet-AI from:

one-time detection

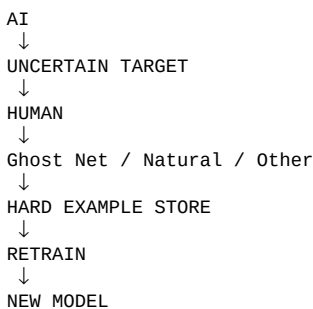
to:

****longitudinal seabed monitoring****

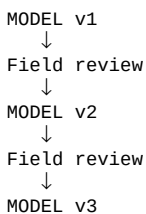
Recent sonar research continues to study change detection across repeat surveys, so this is a legitimate research direction rather than a gimmick. ([Sage Journals](https://journals.sagepub.com/doi/10.1177/02783649241266853?utm_source=chatgpt.com "Machine learning for shipwreck segmentation from side scan sonar imagery: Dataset and benchmark - Advaith V. Sethuraman, Anja Sheppard, Onur Bagoren, Christopher Pinnow, Jamey Anderson, Timothy C. Havens, Katherine A. Skinner, 2025"))

18. Make the system learn from every operator

Your human review can become a research loop:



Then show:



But importantly, don't claim that the model automatically improves unless you actually implement controlled retraining/validation.

The 5 hardest problems I would actually attack

Rather than building every idea above, I would combine them into **five major research modules**:

1. ****Universal Sonar Adaptation****

Handle unseen sonar/environment domains.

2. ****Anthropogenic Evidence Fusion****

Object + shadow + seabed + sonar quality + temporal evidence.

3. ****Open-World / Abstaining Detection****

Known object / unknown anomaly / insufficient evidence.

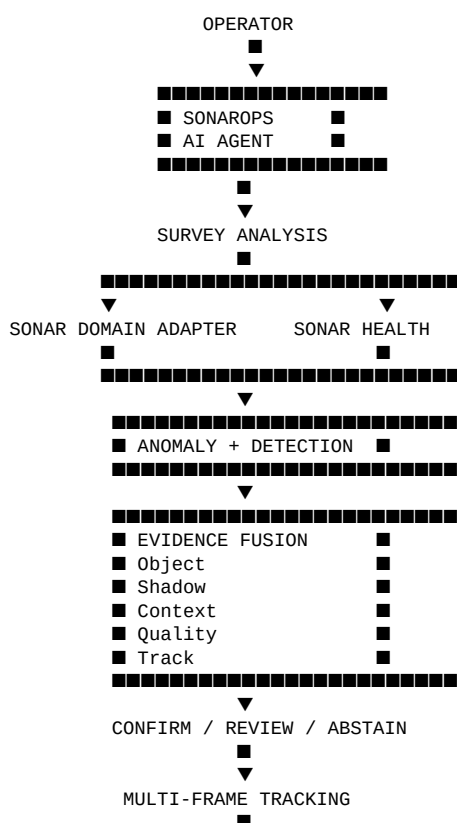
4. ****Track-to-Map Probabilistic Localization****

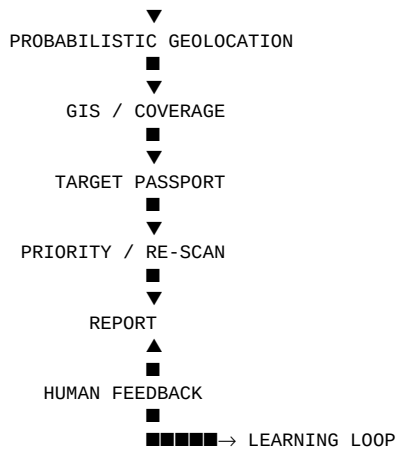
Many sonar observations → one physical target + uncertainty region.

5. ****Autonomous SonarOps Agent****

Agent chooses and executes the processing workflow, while escalating uncertain decisions to humans.

Then your architecture becomes genuinely different





The killer difference

Most teams will likely present:

"We use AI to detect underwater debris."

Your presentation can say:

"GhostNet-AI doesn't only detect an object. It determines whether the evidence is trustworthy, adapts to unfamiliar sonar conditions, consolidates repeated observations into one physical target, estimates where that target actually lies, identifies when the evidence is insufficient, and tells the operator what needs attention next."

That is a much deeper interpretation of the same SIH problem.

And it directly addresses the documented persistent difficulties: **limited labelled sonar data, domain shift, false alarms from natural seabed, changing noise/conditions, small or weak targets, and geolocation uncertainty.** ([Frontiers](https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2025.1542832/full?utm_source=chatgpt.com "Frontiers | Small object detection in side-scan sonar images based on SOCA-YOLO and image restoration"))

One caution: I would **not write "no one has solved this" or "first in the world" in your PPT.** These areas already have active research. Instead, claim the specific architecture and implementation that **your team actually demonstrates.** That is much harder for a jury to challenge.

For the SIH PPT, I would make your new central innovation:

*****"Trustworthy Sonar Intelligence"*****

****Adapt → Detect → Verify → Abstain → Localize → Act****

That is the direction I would build the actual project around.